

Correlation of Fat Free Mass and Fat Mass with Anaerobic Power in Indian Junior Weightlifters

Reet Sarkar¹, Suresh Bhattacharya¹, Anurag Chaurasia²,
Soubhory Ganguly¹, Sourojeet Samanta¹

ABSTRACT

The purpose of the study is to examine the effect of fat free mass and fat mass on the anaerobic power of Indian weightlifters. Subjects for the present study consisted of 06 males and 09 females from SAI NSNIS Patiala, age range between 14 to 21 years. Anaerobic Power was determined by Wingate Test. Fat-Free-Mass and Fat mass was estimated from Bio-Electrical Impedance (BIA). Statistical analysis identified that there is a correlation exists among Anaerobic Power, Fat-Free-Mass and Fat mass. A positive significant correlation existed between anaerobic power and fat-free-mass (0.662) in males & (0.720) for females, while a negative significant correlation existed between anaerobic power and fat mass (-0.744) for males & (-0.052) for females which is non-significant. Anaerobic Power and fat-free-mass are the major factors explaining the explosiveness. It is well established fact that short sprinting events rely heavily upon the anaerobic energy pathways during execution, which leads to speculation that greater performance in a Wingate test can predict success in these events. But on the other hand, additional body fat is detrimental for an athlete or any population as it restricts endurance, flexibility, motion.

KEYWORDS : Anaerobic Power, Fat-Free-Mass, Fat Mass.

INTRODUCTION

Weightlifting is a sport in which barbells are lifted competitively. The origins of modern weightlifting competition are to be found in the 18th and 19th century strong men, such as Eugene Sandow and Arthur Saxon of Germany, George Hackenschmidt of Russia, and Louis Apollon of France, who performed in circuses and theatres. By 1891 there was international competition in London.

The revived Olympic Games of 1896 included weightlifting events, as did the Games of 1900 and 1904.

There are two types of lifting in the weightlifting game - (i) Snatch & (ii) Clean & Jerk. In the snatch the barbell is lifted from the floor to arm's length overhead in a single, continuous, explosive movement, with the lifter being permitted to move his feet or to squat

¹ Department of Sports Physiology, SAI NSNIS, Patiala

² Department of Sports Anthropometry, SAI NSNIS, Patiala

under the barbell as he lifts it before recovering to an erect position. The clean and jerk is a two-part lift. After lifting the barbell to the shoulders, the lifter jerks it overhead to arm's length, with no restrictions on the time necessary to complete the lift or on leg movements. In both lifts, the lifter must complete the lift with feet in line, body erect, arms and legs extended, and the barbell in control overhead. The lifter must either hold the weight overhead for two seconds or wait for the referee's signal before lowering the barbell back to the floor.

As, weightlifting is a power game, it requires lots of anaerobic power and ATP-CP reserve of muscle mass. When movement begins, the breakdown of high-energy molecules known as adenosine triphosphate (ATP) and creatine phosphate (CP) provide immediate energy to power muscle actions. ATP is produced by the breakdown of glucose and glycogen (the storage form of glucose). These molecules can be broken down both with and without oxygen, hence the terms aerobic and anaerobic. The muscular stores of ATP and CP are very limited and become quickly depleted after only a few seconds of activity. At this point, a higher percentage of energy for ATP production is supplied by the glycolytic energy system. These two systems provide energy anaerobically.

Anaerobic power (ANP) is understood as the performance of a work

with maximal speed. The explosive movement needs a fast energy production that occurs with the alactic anaerobic metabolism (ATP-CP). Anaerobic power is important factor in sports with quick, all-out effort. Short sprinting events rely heavily upon the anaerobic energy pathways during execution, which leads to speculation that greater performance in a wingate test can predict success in this event. The present study is mainly concerning about Absolute peak power as well as relative peak power. Absolute peak power [W] - Highest running average of one second in watts. Relative peak power [W/Kg] - Highest running average of one second in Watt per Kg Body weight. Previous Research papers have suggested that there is a strong relationship between anaerobic power and fat free mass. Fat Free Mass (FFM) is the metabolically active tissue in the body. The constituents of FFM are skeletal muscle mass, body cell mass, total body water, and bone mineral mass. The majority of weightlifters are concerned about the amount of body weight because competitors are matched based on body-weight. They generally want to minimize the fat mass level and the total body weight without losing their body strength and power (Yoon, 2002).

METHODS:

The present study was conducted on junior weightlifters (06 males and 09 females) age range between 14 to 21

years which were under training at SAI NSNIS Patiala. At the time of testing all were considered to be physically active as defined by their participating in strenuous physical exercise. Anthropometric parameters, body composition & Physiological tests were administered on two alternative days.

Data Collection:- Anthropometric data collected including standing body height, body weight recorded with the help of SECA Digital BMI Machine (Model No.284)

Measurement of Fat-Free-Mass and Fat mass: FFM & FM was determined by Bio-Electrical Impedance (Body composition analyser Model name: AccunIQ BC 720) method which measure body composition by applying alternative current to athlete to measure their fat mass, fat free mass and other parameter. Analysis was conducted in minimum cloth, and athletes are advised one day before testing that they should not eat, drink and exercise two hours prior to the test.

Physiological data collected including anaerobic power with the help of MONARK Peak Bike (Ergomedic 894 E) and Monark Anaerobic Test Software (Version 3.3.0.0)

Wingate Test for Anaerobic Power: Wingate Test involves 30 seconds of maximal exercise on either an arm-crank or leg cycle ergometer. The testing device is a mechanically-braked cycle ergometer (MONARK). The player followed a five-minute warm-up, which includes three

sprints at varying resistances; the athletes got off the bike during a three-minute recovery or stay on the bike and spin lightly. The athlete then instructed to pedal as fast as possible without any or minimal resistance. Within three seconds, a fixed resistance is applied to the flywheel and the athlete continues to pedal “all out” for the duration of the test (e.g., 30 seconds). A sensor positioned near the flywheel that is interfaced with a computer counted each pedal revolution. The resistance is applied to the flywheel by adding a predetermined amount of weight to the bicycle’s weight tray. The resistance is a percentage of the athlete’s body weight. The resistance load was 7.5% of the subject’s body weight within 0.1kg resolution of resistance range for both male and female.

Statistical Analysis: The means and standard deviations of the anthropometric parameters (age, height, and weight) as well as physiological characteristics were calculated with the help of IBM® SPSS® Statistics [version 28.0.1.0(142)].

Correlations were also computed, using the software packages of the IBM® SPSS® Statistics [version 28.0.1.0 (142)] between all the parameters under study, to elucidate the correlation of anthropometrical and physiological parameters such as FFM, Fat mass and anaerobic power of Indian weightlifters.

RESULTS

Descriptive statistics such as Mean,

Standard Deviation, and Standard Error were calculated between male and female weightlifters from the collected data and are reported for Anthropometry characteristics (height, weight), body

composition (fat mass and fat-free mass and Physiological parameter (Anaerobic Power) can be seen in Tables 1, and 2 respectively.

Table – 1: Descriptive Statistics of Male and Female weightlifters (N=15)

Statistics	Gender	Height (cm)	Weight (Kg)	BMI (Kg/m ²)	Fat Free Mass (Kg)	Body Fat(kg)	Relative Peak Power (W/Kg)
Mean	Male (n=6)	171.45	93.66	31.9	68.6	25.06	10.88
Standard Deviation		±5.79	±6.95	±2.36	±8.10	±4.20	±1.27
Standard Error		2.36	2.84	0.96	3.30	1.71	0.52
Mean	Female (n=9)	160.24	65.85	25.44	45.40	20.45	7.95
Standard Deviation		±6.18	±14.61	±4.19	±6.88	±8.45	±1.16
Standard Error		2.06	4.87	1.39	2.29	2.81	0.38

Table-1 shows the physical data of the weightlifters. Age range between 14 to 21 was measured. Mean Height of male and female subjects were 171.45 cm (±5.79) and 160.24cm (±6.18) respectively. Mean Weight was measured 93.67kg (±6.95) in males and 65.85kg (±14.61) in females. Mean BMI of Male and Female subjects were measured 31.9 (±2.36) and 25.44 (±4.19) respectively. Mean Fat-Free-Mass of Male and Female subjects were measured 68.6 kg and 45.40 kg respectively while the fat mass of both the subjects were measured 25.06 kg and

20.45 kg respectively. Mean Relative Peak Power was measured 10.88 (W/Kg) in Males and 7.95 (W/Kg) in Females. Performance levels can be measured by looking at the power output. There have been several studies describing the power output of males (Garhammer, 1979; Garhammer, 1980; and Garhammer, 1985). Garhammer (1991) later published power output data on world class female athletes who competed in the first Women's World Weightlifting Championships.

Table-2: Correlation between Anaerobic Power and Fat-Free Mass of Weightlifters

Variable	Gender	Corelation coefficient	Sig.
Anaerobic Power & Fat Free Mass	Male	0.152	0.662*
	Female	0.855	0.720*

*Significant at 5% level (0.62; n=6)

Table-2 shows the correlation between Anaerobic Power and Fat-Free-Mass of Weightlifters. From the table, it indicates that there is a positive correlation (0.662 for male and 0.720 for female) existed. As

positive correlation exists between Anaerobic Power and Fat-Free-Mass, it can be easily said from here that if Fat-Free Mass increases, the anaerobic power will increase.

Table – 3: Correlation between Anaerobic Power and Fat Mass of Weightlifters

Variable	Gender	Corelation coefficient	Sig.
Anaerobic Power & Fat mass	Male	- 0.090	- 0.744
	Female	- 0.895	- 0.052

Table - 3 shows the correlation between anaerobic power and fat mass of athletes, depicts that there is a negative correlation (-0.744 for male; -0.052 for female) between the two parameters. As negative correlation exists between Anaerobic Power and Fat mass, it can be interpreted that if Fat mass increases, the anaerobic power will decrease.

DISCUSSION:

FFM and Anaerobic Power are very important in weightlifters for their performance in competitions because they theoretically explain the skeletal

muscle mass of the body, which has a substantial effect on the weightlifters' physical strength. For the Weightlifters' continuous and repetitive efforts during a very short time frame is required. Anaerobic capacity is regarded as the most important physical condition for effective technique performance. Energy contribution in the maximum exertion of strength for 30 seconds and the anaerobic energy contribution ratio is 72% (Adenosine Tri-Phosphate [ATP] - Phosphocreatine [PCr] system: 23%, Glycolysis: 49%).

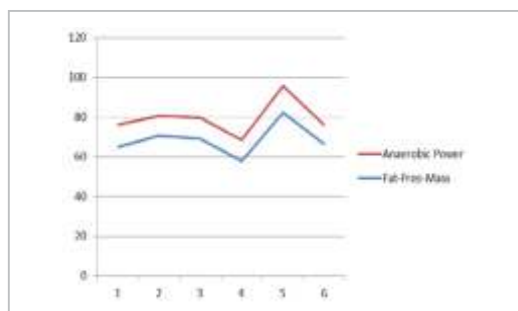


Fig 1: Represents the relationship between anaerobic power and FFM in Males

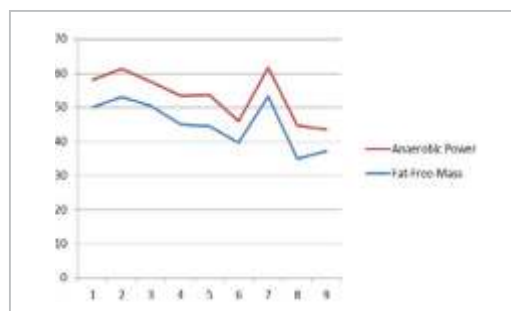


Fig 2: Represents the relationship between anaerobic power and FFM in Females

Peak power represents the highest amount of power generated during the test; it is usually achieved within the first five seconds. This value indicates the energy generating capacity of the ATP-CP system, and it is measured in watts. Relative power represents peak power divided by the athlete's body mass. Reporting data in relative units is actually of more valuable than absolute units when making comparisons between athletes. It has been suggested that anaerobic capacity plays a very important role in Weightlifters' activities during a competition. Because the anaerobic energy contribution ratio of 72% is highly dependent on PCr in skeletal muscle, it is directly affected by FFM.

These significant differences might come from the difference in FFM of body composition. It is believed that there were tendency differences among the groups because FFM contain muscle fibres indispensable for the contraction and recovery of muscle. Based on the results of this study, we found that weightlifters' FFM are directly correlated with their peak power and show tendency differences among groups. Fat-free mass showed a very high correlation with peak power (Male: $r = 0.662$; Female: $r = 0.720$ $p < 0.05$). This suggests that to improve anaerobic power, need to increase FFM.

Other findings of the present study were: female weightlifters were shorter, presented lower fat-free mass, and higher fat mass compared to males, all with

moderate effect sizes; peak and mean power were higher for males compared to females, with moderate effect sizes; significant, large and very large, positive correlations were found between Wingate and anaerobic power. The higher Wingate peak power in male compared to female weightlifters have been reported previously, and typically females present approximately 70% of those observed in males [Franchini et al. 2011]. This difference is attributed to higher glycolytic participation (35%) during the Wingate test in males compared to females, and higher aerobic contribution (7%) in females compared to males [Hill, Smith 1993]. Along with above discussion, anaerobic power also depends on two factors – (i) Type of muscle fibres, (ii) ATP-CP Reserve. It can be said that if an athlete has higher ATP-CP Reserve, the athlete's anaerobic power should be more. It is due to the fact that if an athlete has more ATP-CP Reserve, the athlete can use that reserve during the 30s second test or in a short-time event or competition where explosiveness is tested. So, athlete can rely on athlete's ATP-CP reserve not on the aerobic glycolysis too much. Among the two factors, the type of muscle fibres (the First one) is one of the matter concerns in case of anaerobic power. Previous research papers have suggested that, Type IIb fibres are mainly associated with explosiveness or anaerobic power whereas Type I (Slow) and Type IIa (Fast)

fibres are associated with endurance activity. Type IIb fibres are mainly associated with anaerobic power just because of the pathway it is used for ATP production. The pathway Type IIb fibres use is anaerobic glycolysis which cannot be sustained for as long as a period of time as aerobic cellular respiration (the main pathway for ATP production in Type I and Type IIa). As Type IIb fibres use anaerobic glycolysis for ATP production, it has the least resistance to fatigue.

According to Cooper (1968), higher fat percentage causes lower performance in activities which involve body

displacement. The excess fat mass can interfere directly with the weightlifter's performance (jerk) due to increase in body weight, which diminishes the athlete's acceleration, since acceleration is equal to force divided by mass. This indicates that weightlifters with moderate values of fat mass may be able to perform better as compared to those with higher fat mass values. Thus, gradual and reasonable individual programs of fat mass reduction with professional guidance should be recommended in weightlifters to improve their physical fitness.

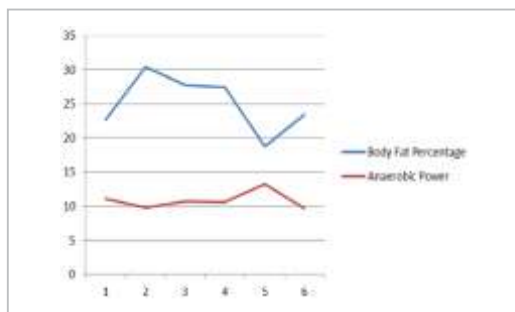


Fig 3 represents the relationship between anaerobic power and % of Body Fat in Males.

The additional fat mass can limit endurance, balance, coordination, and movement capacity. Joint range of motion can be negatively affected by excessive body mass and fat as well and mass can form a physical barrier to joint movement in a complete range of motion. Thus, athletes competing in sports that require high levels of flexibility benefit from having low levels of body fat.

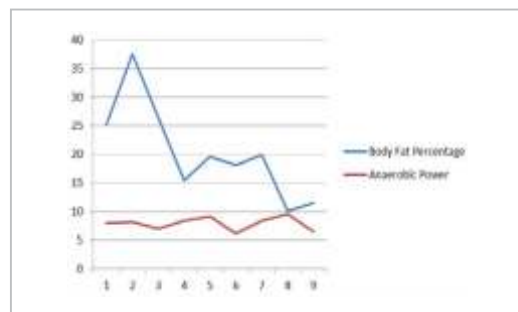


Fig 4 represents the relationship between anaerobic power and % of Body Fat in Females.

All fitness components depend on body composition to some extent. An increase in fat mass contributes to strength and power development. Strength and power are related to muscle size. Thus, an increase in lean FFM enables the athlete to generate more force in a specific period of time. A sufficient level of lean body mass also contributes to speed, quickness, and agility

performance (in the development of force applied to the ground for maximal acceleration and deceleration). Reduced nonessential fat mass contributes to muscular and cardiorespiratory endurance, speed, and agility development. Additional weight (in the form of nonessential fat) provides greater resistance to athletic motion thereby forcing the athlete to increase the muscle force of contraction per given workload.

CONCLUSION

As, weightlifting is a power game; it requires lots of anaerobic power and ATP-CP reserve of muscle mass. Anaerobic power depends on Fat free mass of an athlete. We can predict that an athlete, who possesses more fat free mass, should have more anaerobic power. To sum up the present study showed there is a strong positive correlation of anaerobic power with fat free mass (0.662 in males and 0.720 in females). It might be because of, (i) Energy contribution in the maximum exertion of strength for 30 seconds and the anaerobic energy contribution ratio is

72% (Adenosine Tri-Phosphate [ATP] - Phosphocreatine [PCr] system: 23%, Glycolysis: 49%), (ii) Type of muscle fibers & (iii) ATP-CP Reserve.

A negative correlation with fat mass (-0.744 in males and -0.052 in females). It might be because of, (i) higher fat mass causes lower performance in activities which involve body displacement, (ii) the additional fat mass can limit endurance, balance, coordination and movement capacity. Joint range of motion can be negatively affected by excessive body mass and fat as well and mass can form a physical barrier to joint movement in a complete range of motion.

Further studies are required to understand the correlation of FFM, Fat mass and Anaerobic Power with large number of subjects so that the data error will minimize and makes it more convenient. As, Indian population is more prone to fat mass and less fat free mass, so sports scientist should vigilant about the study among anaerobic power, Fat free mass and fat mass.

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